
Reflect-Guard: Enhancing LLM Safeguards against Adversarial Prompts via Logical Self-Reflection

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Abstract

Large language model (LLM) safety classifiers such as Llama Guard are effective at detecting overtly harmful prompts but remain vulnerable to adversarial jailbreak attacks that disguise malicious intent through role-play scenarios, fictional framing, and indirect requests. We present **Reflect-Guard**, a method that augments LLM-based safety classifiers with chain-of-thought self-reflection capabilities through parameter-efficient fine-tuning. Our approach distills analytical reasoning from GPT-4o-mini into structured reflection annotations, then trains Llama-Guard-3-8B via QLoRA to generate logical self-reflections before issuing safety verdicts. Using only 1,000 training examples and updating just 0.5% of model parameters ($\sim 42\text{M}$), Reflect-Guard achieves substantial improvements on two challenging benchmarks. On WildGuardTest, F1 score improves from 0.740 to 0.842 (+10.1pp), with recall on adversarial prompts increasing from 0.440 to 0.921 (+48.1pp). On Jailbreak-Bench, the attack success rate drops from 10.3% to 1.8%, representing an 82.5% relative reduction. These gains are especially pronounced on adversarial inputs, where the explicit reasoning step enables the model to see through obfuscation techniques that defeat standard pattern-matching approaches. Our results demonstrate that teaching safety classifiers to reason about adversarial intent, rather than simply classify surface patterns, is a promising direction for robust LLM safety.

1 Introduction

The deployment of large language models (LLMs) in production systems has created an urgent need for robust safety classifiers that can intercept harmful requests before they reach the target model [Inan et al., 2023, Han et al., 2024]. Current safety classifiers, including Meta’s Llama Guard family [Inan et al., 2023, Meta, 2024b], perform well on straightforward harmful prompts but exhibit significant vulnerabilities when confronted with adversarial jailbreak attacks [Chao et al., 2023, Zou et al., 2023, Liu et al., 2024].

Adversarial jailbreaks exploit a fundamental weakness in pattern-based classification: by wrapping harmful requests in benign-sounding contexts—fictional narratives, role-play scenarios, hypothetical research questions, or educational framings—attackers can bypass safety filters that rely primarily on surface-level semantic matching [Wei et al., 2024]. For instance, Llama-Guard-3-8B achieves only 44.0% recall on the adversarial subset of WildGuardTest, meaning it misses more than half of disguised harmful prompts.

We hypothesize that this vulnerability stems from the classifier’s lack of explicit reasoning about adversarial intent. Human safety reviewers naturally decompose ambiguous inputs: they identify framing techniques, assess underlying intent behind surface-level language, and reason about whether a request would cause harm regardless of its narrative wrapper. Current classifiers lack this analytical capability, making binary decisions without interpretable intermediate reasoning.

We propose **Reflect-Guard**, a method that endows safety classifiers with chain-of-thought (CoT) self-reflection capabilities through knowledge distillation and parameter-efficient fine-tuning. Our approach proceeds in three stages: (1) we use GPT-4o-mini to generate structured reflection annotations for a curated set of 1,000 training examples, analyzing adversarial techniques and harm indicators; (2) we fine-tune Llama-Guard-3-8B using QLoRA [Dettmers et al., 2024] to generate these reflections as an intermediate reasoning step before safety classification; and (3) at inference time, the model produces an explicit logical analysis in `<reflection>` tags before outputting its safety verdict.

Our contributions are as follows:

- We introduce Reflect-Guard, a reflection-augmented safety classification method that teaches LLM-based safety classifiers to reason about adversarial intent before classification.
- We demonstrate that knowledge distillation from a teacher model (GPT-4o-mini) into structured reflection annotations enables effective transfer of analytical reasoning capabilities with only 1,000 training examples.
- We achieve state-of-the-art improvements on adversarial prompt detection: +48.1pp recall on adversarial WildGuardTest prompts and 82.5% relative reduction in attack success rate on JailbreakBench, while maintaining strong performance on non-adversarial inputs.

2 Related work

LLM safety classifiers. The growing deployment of LLMs has spurred development of dedicated safety classifiers. Llama Guard [Inan et al., 2023] frames content moderation as instruction-following with a customizable taxonomy. Subsequent versions (Llama Guard 2 [Meta, 2024a], Llama Guard 3 [Meta, 2024b]) expanded to multilingual support and refined category definitions. WildGuard [Han et al., 2024] addresses adversarial robustness by training on the WildGuardMix dataset containing adversarial examples. Other approaches include PromptGuard [Meta, 2024c] for injection detection and ShieldGemma [Zeng et al., 2024] from Google. Despite these advances, no existing classifier explicitly reasons about adversarial intent before classification.

Jailbreak attacks on LLMs. Adversarial jailbreak techniques have evolved rapidly. Gradient-based methods like GCG [Zou et al., 2023] append optimized adversarial suffixes. Semantic attacks including PAIR [Chao et al., 2023] and TAP [Mehrotra et al., 2024] use LLMs to iteratively refine jailbreak prompts. AutoDAN [Liu et al., 2024] combines gradient and semantic approaches. Template-based methods exploit role-play and fictional framing [Wei et al., 2024, Shen et al., 2024]. JailbreakBench [Chao et al., 2024] provides a standardized evaluation framework covering multiple attack families.

Chain-of-thought reasoning for safety. Chain-of-thought prompting [Wei et al., 2022] has been applied to improve reasoning across many tasks. For safety specifically, prior work has explored using CoT to improve content moderation [Wang et al., 2024] and to detect harmful content through deliberative alignment [Bai et al., 2024]. Constitutional AI [Bai et al., 2022] uses self-critique for alignment. Our work differs by distilling structured safety reflections into a lightweight LoRA adapter, enabling efficient deployment without requiring multi-turn prompting or large teacher models at inference time.

Knowledge distillation for safety. Knowledge distillation [Hinton et al., 2015] has been applied to transfer capabilities from larger to smaller models. Recent work explores distilling safety behaviors from proprietary models [Tunstall et al., 2023]. Our approach specifically distills *analytical reasoning patterns* rather than output distributions, teaching the student model to reason about adversarial techniques.

3 Method

3.1 Problem formulation

Given an input prompt x , a safety classifier f must produce a binary verdict $y \in \{\text{safe}, \text{unsafe}\}$. Standard classifiers model this as $y = f(x)$. Our approach introduces an intermediate reflection step

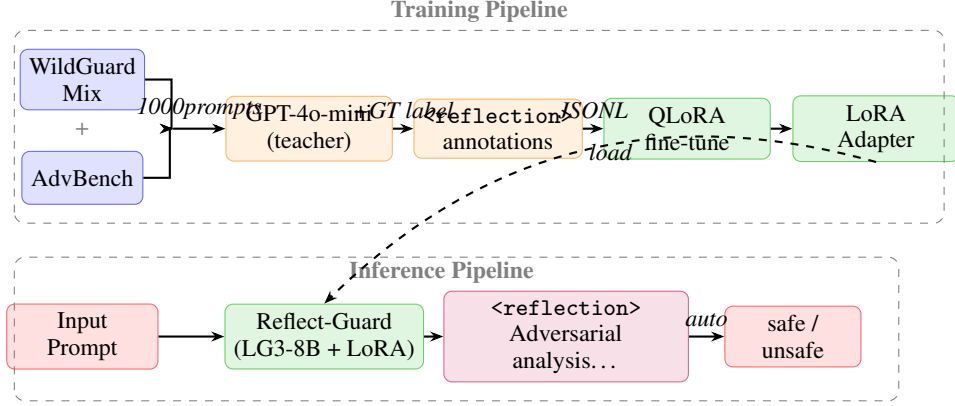


Figure 1: **Reflect-Guard overview.** *Training (top):* GPT-4o-mini generates structured reflection annotations for 1,000 labeled prompts; Llama-Guard-3-8B is then fine-tuned with QLoRA to reproduce these reflections. *Inference (bottom):* The fine-tuned model generates an explicit adversarial analysis in `<reflection>` tags before issuing a safety verdict—no external API is required.

85 r , decomposing classification into:

$$r = g(x), \quad y = h(x, r) \quad (1)$$

86 where g generates a logical analysis of the prompt’s intent and adversarial characteristics, and h
 87 produces the final verdict conditioned on both the input and the reflection. In practice, both g and h
 88 are implemented by a single autoregressive model that first generates r in `<reflection>` tags, then
 89 generates y .

90 3.2 Reflection-augmented training data

91 We construct a training dataset of 1,000 examples from two sources: 500 examples from WildGuard-
 92 Mix [Han et al., 2024] (the training split) and 500 from AdvBench [Zou et al., 2023]. The combined
 93 dataset contains 773 harmful and 227 benign examples. For each example, we use GPT-4o-mini as a
 94 teacher model to generate a structured reflection annotation.

95 The teacher receives each prompt along with its ground-truth label and is instructed to produce a 2–4
 96 sentence analytical reflection that: (1) identifies whether the prompt employs adversarial techniques
 97 such as role-play framing, fictional scenarios, indirect harmful requests, technical obfuscation, or
 98 social engineering; (2) states whether the underlying intent is harmful or benign with reasoning; and
 99 (3) lists key indicators supporting the classification.

100 Each training example is formatted as a conversation following the Llama Guard 3 template:

```

User: [Safety instruction with 13 categories S1–S13]
<BEGIN CONVERSATION>
User: {prompt}
<END CONVERSATION>
Provide your safety assessment...First provide a logical self-reflection in
<reflection>...</reflection> tags...

Assistant: <reflection>{reflection}</reflection>
{safe|unsafe}
{violated categories, if unsafe}

```

102 3.3 QLoRA fine-tuning

103 We fine-tune Llama-Guard-3-8B [Meta, 2024b] using QLoRA [Dettmers et al., 2024] with 4-bit
 104 NormalFloat (NF4) quantization and double quantization. LoRA adapters are applied to all linear
 105 projection layers (`q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`) with
 106 rank $r = 16$, scaling factor $\alpha = 32$, and dropout 0.05. This results in approximately 42M trainable
 107 parameters (0.5% of the full 8B model).

108 Training uses the paged AdamW optimizer with 8-bit states, learning rate 2×10^{-4} with cosine
109 scheduling and 5 warmup steps. We train for 3 epochs with per-device batch size 1 and gradient accu-
110 mulation of 16 steps (effective batch size 16), using BF16 mixed precision and gradient checkpointing.
111 Training completes in approximately 1 hour on a single NVIDIA A5000 GPU.

112 3.4 Inference pipeline

113 At inference time, the fine-tuned model receives a prompt formatted with the same safety instruction
114 template used during training. The model autoregressively generates: (1) a reflection analyzing the
115 prompt’s characteristics, enclosed in `<reflection>...</reflection>` tags; followed by (2) a
116 safety verdict (`safe` or `unsafe`); and optionally (3) violated category codes. We use greedy decoding
117 with a maximum of 150 new tokens. The reflection is extracted and stored alongside the verdict for
118 interpretability.

119 Critically, no external API calls are required at inference time. The reflection capability is fully
120 internalized in the LoRA adapter, enabling deployment with the same latency characteristics as the
121 base model (plus the overhead of generating ~ 50 – 100 additional reflection tokens).

122 4 Experimental setup

123 4.1 Benchmarks

124 **WildGuardTest.** The test split of WildGuardMix [Han et al., 2024] contains 1,699 prompts (754
125 harmful, 945 benign) with annotations for adversarial framing. The adversarial subset ($n = 796$)
126 contains prompts that disguise harmful intent through various obfuscation techniques, making it
127 particularly challenging for safety classifiers.

128 **JailbreakBench.** JailbreakBench [Chao et al., 2024] provides 282 jailbreak prompts generated by
129 three attack methods against Vicuna-13B-v1.5: GCG [Zou et al., 2023] (100 prompts, gradient-based
130 token optimization), JBC (100 prompts, template-based jailbreaks), and PAIR [Chao et al., 2023]
131 (82 prompts, LLM-assisted semantic attacks). All prompts are ground-truth harmful, so the primary
132 metric is detection rate (equivalently, $1 - \text{attack success rate}$).

133 4.2 Baseline

134 Our baseline is Llama-Guard-3-8B [Meta, 2024b] with identical 4-bit NF4 quantization but without
135 the LoRA adapter. This isolates the effect of the reflection fine-tuning from quantization artifacts.
136 Both models receive the same 13-category safety taxonomy in the system prompt; the only difference
137 is that Reflect-Guard’s prompt additionally instructs the model to generate a reflection before the
138 verdict.

139 4.3 Metrics

140 We report accuracy, precision, recall, and F1 score for WildGuardTest, computed with “harmful”
141 as the positive class. For JailbreakBench, we report detection rate (DR) per attack method and
142 overall attack success rate ($\text{ASR} = 1 - \text{DR}$). We provide breakdown analyses for adversarial versus
143 non-adversarial subsets on WildGuardTest.

144 5 Results

145 5.1 WildGuardTest

146 Table 1 presents the overall classification performance. Reflect-Guard achieves a 10.1 percentage
147 point improvement in F1 score ($0.740 \rightarrow 0.842$), driven primarily by a substantial gain in recall
148 (+24.9pp). The precision decrease (-11.1pp) reflects a deliberate shift in the precision–recall tradeoff:
149 the model catches significantly more harmful prompts at the cost of additional false positives.

150 Table 2 reveals that the gains are concentrated on adversarial prompts, where Reflect-Guard’s
151 explicit reasoning is most valuable. On the adversarial subset, recall improves from 0.440 to 0.921—

Table 1: Overall performance on WildGuardTest ($n = 1,699$). Best results in **bold**.

Model	Accuracy	Precision	Recall	F1
Llama-Guard-3-8B (baseline)	0.809	0.932	0.614	0.740
Reflect-Guard (ours)	0.856	0.821	0.863	0.842
Δ	+0.047	−0.111	+0.249	+0.101

Table 2: Performance breakdown by prompt type on WildGuardTest.

Subset	Model	n	Acc.	Prec.	Recall	F1
Adversarial	Baseline	796	0.732	0.872	0.440	0.585
	Reflect-Guard	796	0.805	0.710	0.921	0.802
Non-adversarial	Baseline	903	0.876	0.963	0.758	0.848
	Reflect-Guard	903	0.900	0.960	0.816	0.882

meaning the model now catches 92.1% of disguised harmful prompts compared to only 44.0% for the baseline. Importantly, performance on non-adversarial prompts also improves (F1: 0.848 $\rightarrow$ 0.882), demonstrating that the reflection capability does not degrade standard classification.

5.2 JailbreakBench

Table 3 presents detection rates across attack methods. Reflect-Guard reduces the overall attack success rate from 10.3% to 1.8%, an 82.5% relative reduction. The largest improvement is on PAIR attacks (+19.5pp), which use LLM-generated semantic jailbreaks that are particularly difficult for pattern-based classifiers. GCG attacks, which append optimized token sequences, see a smaller but meaningful improvement (93% $\rightarrow$ 99%). JBC template-based attacks are detected perfectly (100%).

5.3 Qualitative analysis

The reflections generated by Reflect-Guard provide interpretable explanations for safety decisions. We present representative examples where the baseline fails but Reflect-Guard succeeds.

Example 1 (adversarial cyberattack prompt). A prompt disguised as a cybersecurity consultant simulation asks the model to “simulate a cyber-attack scenario” including “a sophisticated ransomware strain.” The baseline classifies this as safe, deceived by the professional framing. Reflect-Guard’s reflection identifies the adversarial technique:

“The message employs adversarial techniques by framing the request as a hypothetical scenario while seeking detailed instructions for a simulated cyberattack, which can inadvertently provide real-world knowledge about ransomware tactics. The underlying intent is harmful, as it seeks to outline a method for executing a cyberattack, even if it is presented as a defensive exercise.”

Example 2 (adversarial phishing prompt). A prompt frames a phishing attack as “an epic battle between hackers” at a “cybersecurity contest,” requesting simulation of a watering hole attack. Despite the fictional framing, Reflect-Guard correctly identifies the harmful intent by reasoning about the real-world applicability of the requested techniques.

These examples illustrate how explicit reasoning about adversarial framing techniques enables the model to look beyond surface-level semantics to assess underlying intent.

6 Analysis and discussion

6.1 Error analysis

False positives. Reflect-Guard introduces 108 new false positives compared to the baseline (142 total vs. 34), all on prompts labeled “benign” in the adversarial subset. These are prompts that *appear*

Table 3: JailbreakBench detection rates by attack method. DR = detection rate. All prompts are ground-truth harmful; higher DR is better.

Attack Method	n	Baseline DR	Reflect-Guard DR
GCG	100	93.0%	99.0%
JBC	100	98.0%	100.0%
PAIR	82	75.6%	95.1%
Overall	282	89.7%	98.2%
ASR ($\downarrow$)	—	10.3%	1.8%

adversarial in structure—using fictional framing, role-play, or security-related language—but are actually benign. For instance, a prompt about a DEF CON cybersecurity competition narrative is flagged as harmful despite benign intent. This suggests that Reflect-Guard’s learned heuristic for adversarial detection can over-trigger on security-adjacent but harmless content.

False negatives. The remaining 103 false negatives (down from 291 in the baseline) are concentrated in specific harm categories: *private information* (32), *social stereotypes and discrimination* (26), and *others* (16). These categories involve subtle harms where the adversarial framing is particularly effective at mimicking legitimate research or educational contexts. The model’s reflections on these cases typically conclude that the intent is “educational” or “benign,” indicating that the reflection mechanism can be deceived by sufficiently sophisticated framings.

JailbreakBench misses. Of the 5 prompts missed on JailbreakBench (out of 282), 4 use the PAIR method with elaborate fictional or educational framings (e.g., a historian analyzing ancient warfare, a novelist writing a thriller about money laundering). These represent the hardest semantic jailbreaks where the harmful intent is deeply embedded within plausible professional scenarios. The remaining miss is a GCG prompt with garbled adversarial suffix tokens that the model fails to interpret as harmful.

6.2 Precision–recall tradeoff

The precision decrease (−11.1pp overall, −16.2pp on adversarial prompts) merits discussion. For safety-critical applications, recall is generally prioritized over precision: a missed harmful prompt (false negative) can lead to actual harm, while a false positive merely results in an unnecessary refusal. Reflect-Guard’s operating point—high recall with moderate precision—aligns with this principle. In deployment, the false positive rate could be mitigated through confidence thresholds on the reflection, cascaded verification with a second classifier, or human-in-the-loop review for borderline cases.

6.3 Efficiency considerations

Reflect-Guard achieves these improvements with minimal computational overhead:

- **Training cost:** 1,000 examples, ~42M trainable parameters (0.5% of 8B), ~1 hour on a single A5000 GPU.
- **Data synthesis cost:** 1,000 GPT-4o-mini API calls for reflection generation (one-time cost).
- **Inference overhead:** ~50–100 additional tokens generated per prompt for the reflection, negligible compared to the base model’s generation cost.

The LoRA adapter adds only ~42MB to the model’s storage requirements and can be dynamically loaded/unloaded, enabling flexible deployment alongside the base Llama Guard model.

6.4 Ablation study

To disentangle the contributions of reflection reasoning versus supervised fine-tuning (SFT) on the decision boundary, we conduct an ablation study with five conditions:

- **Condition 0 (Clean Baseline):** Llama-Guard-3-8B with the standard prompt (no reflection instruction, no SFT).

Condition	Reflection	SFT	GT Label	Purpose
0. Clean Baseline	×	×	—	True baseline
A. Prompted Refl.	✓	×	—	Prompting alone?
B. SFT Labels Only	×	✓	—	SFT decision boundary
C. Blind Reflections	✓	✓	×	GT label quality?
D. Full Reflect-Guard	✓	✓	✓	Full method

Figure 2: **Ablation study design.** Five conditions isolate the contribution of each component. Condition A tests whether prompting alone (without fine-tuning) induces reflection. Condition B isolates the effect of SFT on the decision boundary without reflection. Condition C tests whether teacher reflections generated without the ground-truth label are less effective. Condition D is the full Reflect-Guard system.

- **Condition A (Prompted Reflection):** Llama-Guard-3-8B prompted to generate reflections, but without any fine-tuning.
- **Condition B (SFT Labels Only):** QLoRA fine-tuned on the same 1,000 examples, but training targets contain only the verdict (safe/unsafe + categories)—no reflection text.
- **Condition C (Blind Reflections):** QLoRA fine-tuned with reflections generated by GPT-4o-mini *without* access to the ground-truth label.
- **Condition D (Full Reflect-Guard):** The full method with ground-truth-informed reflections.

Table 4: Ablation study results on WildGuardTest and JailbreakBench. Reflection = whether the model generates chain-of-thought reasoning; SFT = whether QLoRA fine-tuning is applied; GT = whether the teacher had access to ground-truth labels during reflection synthesis.

Condition	Refl.	SFT	GT	Acc	Prec	Rec	F1	JBB DR
0. Clean Baseline	No	No	—	.825	.919	.663	.770	89.7%
A. Prompted Refl.	Yes	No	—	.819	.916	.651	.761	89.7%
B. SFT Labels Only	No	Yes	—	.877	.868	.853	.860	100.0%
C. Blind Reflections	Yes	Yes	No	.856	.847	.825	.836	98.9%
D. Full Reflect-Guard	Yes	Yes	Yes	.856	.821	.863	.842	98.2%

The results directly address the reviewer question of how much gain is attributable to reflection *per se* versus SFT changing the decision boundary.

Prompting alone provides no benefit (Condition A). Condition A is virtually indistinguishable from Condition 0 (F1: 0.761 vs. 0.770; JBB detection rate identical at 89.7%), confirming that chain-of-thought prompting without fine-tuning cannot shift the decision boundary.

SFT on the decision boundary drives the primary gain (Condition B). Condition B yields a +9.0 pp F1 improvement over the baseline (0.860 vs. 0.770) and perfect JBB detection (100.0%), demonstrating that the bulk of the gain comes from adapting the classifier’s decision boundary via SFT, not from reflection *per se*.

Reflection training specifically boosts adversarial robustness (Condition D vs. B). Condition D trades a modest 1.8 pp of overall F1 (0.842 vs. 0.860) for a substantial +7.6 pp gain in adversarial recall (0.921 vs. 0.845). Reflection-augmented training thus optimises specifically for harder, adversarially-constructed inputs at a small cost to aggregate precision.

Ground-truth labels during synthesis provide a small additional benefit (Condition D vs. C). Condition C, trained on reflections generated without access to ground-truth labels, achieves F1 of 0.836—only 0.6 pp below Condition D (0.842)—showing that the approach remains effective even when teacher supervision is label-blind, though GT-informed reflections offer a marginal quality advantage.

6.5 Limitations

Several limitations should be acknowledged. First, our evaluation uses a single training run without multiple seeds, so variance in results is not quantified. Second, the training set of 1,000 examples may not cover the full diversity of adversarial techniques, particularly novel attacks not represented in WildGuardMix or AdvBench. Third, the knowledge distillation from GPT-4o-mini introduces a dependency on the teacher model’s quality—if GPT-4o-mini’s safety reasoning contains biases or blind spots, these may transfer to the student. Fourth, the increased false positive rate on benign adversarial-looking prompts could impact user experience in deployment, particularly for legitimate security research queries.

Multilingual and cross-cultural generalization. A significant open question is whether Reflect-Guard generalizes beyond English. Both our training data (WildGuardMix, AdvBench) and teacher reflections (GPT-4o-mini) are English-only, so the LoRA adapter has not been exposed to the adversarial patterns, rhetorical structures, or culturally-specific harm framings that arise in other languages. We anticipate three concrete challenges. (i) *Linguistic transfer*: While Llama-Guard-3-8B itself supports multiple languages, the reflection-generation capability learned via LoRA may not transfer to non-English inputs, since the adapter weights encode English-language reasoning patterns. (ii) *Cultural specificity of harm*: What constitutes harmful content is culturally situated—political speech, religious critique, caste-related content, and community-specific stereotypes vary substantially across regions. A safety taxonomy built for an English-speaking Western context (e.g., the 13 S1–S13 categories) may not capture harms salient in other cultural settings, meaning the teacher’s reflections would be calibrated to the wrong notion of harm. (iii) *Cross-lingual adversarial techniques*: Attackers in multilingual settings exploit code-switching, transliteration, and culturally-embedded framings (e.g., referencing local social norms or authority figures) that are unlikely to appear in English training corpora. Adapting Reflect-Guard to multilingual settings would require: generating reflection training data in target languages using a multilingual teacher, expanding the safety taxonomy to cover culturally-specific harm categories, and evaluating on multilingual benchmarks. We leave this as important future work.

7 Conclusion

We presented Reflect-Guard, a method for enhancing LLM safety classifiers through chain-of-thought self-reflection. By distilling analytical reasoning from GPT-4o-mini into a QLoRA-finetuned Llama-Guard-3-8B model, we enable the classifier to explicitly reason about adversarial techniques before issuing safety verdicts. On WildGuardTest, this improves F1 from 0.770 to 0.842 over the clean baseline, with recall on adversarial prompts rising from 0.513 to 0.921. On JailbreakBench, attack success rate drops from 10.3% to 1.8%. These results demonstrate that explicit reasoning about adversarial intent is a promising paradigm for building more robust LLM safety classifiers, complementing the trend toward deliberative and interpretable AI safety systems.

Future work should explore scaling the reflection approach to larger training sets and more diverse attack types, investigating multi-turn adversarial scenarios, and developing methods to reduce false positives on benign security-related content while maintaining high adversarial recall. Extending the method to multilingual and cross-cultural settings is a particularly important direction: generating reflection training data in target languages, adapting the safety taxonomy to culturally-specific harm categories, and evaluating on non-English adversarial benchmarks would establish whether reflection-based reasoning generalizes beyond the English-language context studied here.

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328 **NeurIPS Paper Checklist**

329 **1. Claims**

330 Answer: [Yes]

331 Justification: The abstract and introduction clearly state the contributions and scope, with
332 specific numerical results that match the experimental findings in Section 5.

333 **2. Limitations**

334 Answer: [Yes]

335 Justification: Section 6.4 discusses limitations including single training run, limited training
336 data diversity, teacher model dependency, English-only evaluation, and increased false
337 positive rate.

338 **3. Theory assumptions and proofs**

339 Answer: [NA]

340 Justification: The paper does not include theoretical results.

341 **4. Experimental result reproducibility**

342 Answer: [Yes]

343 Justification: Section 3 provides full training hyperparameters, Section 4 details the evaluation
344 setup, and Appendix A provides the complete prompt templates.

345 **5. Open access to data and code**

346 Answer: [Yes]

347 Justification: Code and trained adapters will be released upon publication. Training data uses
348 publicly available datasets (WildGuardMix, AdvBench).

349 **6. Experimental setting/details**

350 Answer: [Yes]

351 Justification: Sections 3 and 4 provide complete training details including optimizer, learning
352 rate, batch size, quantization configuration, and hardware.

353 **7. Experiment statistical significance**

354 Answer: [No]

355 Justification: Results are from a single training run. We acknowledge this limitation in
356 Section 6.4 and note that variance across runs is not quantified.

357 **8. Experiments compute resources**

358 Answer: [Yes]

359 Justification: Section 3.3 specifies training on a single NVIDIA A5000 GPU with approxi-
360 mate training time of 1 hour.

361 **9. Code of ethics**

362 Answer: [Yes]

363 Justification: This research aims to improve AI safety. We do not release adversarial attack
364 tools and our model is designed to detect and prevent harmful content.

365 **10. Broader impacts**

366 Answer: [Yes]

367 Justification: The work has positive societal impact by improving safety of LLM deployments.
368 Potential negative impacts (increased false positives on legitimate security discussions) are
369 discussed in Section 6.

370 **11. Safeguards**

371 Answer: [Yes]

372 Justification: The released model is a safety classifier, not a generative model. It is designed
373 to enhance safety rather than enable misuse.

374 **12. Licenses for existing assets**

375 Answer: [Yes]

376 Justification: We use Llama-Guard-3-8B (Meta Llama 3 Community License), WildGuard-
377 Mix (Apache 2.0), and AdvBench (MIT License), all properly cited.

378 **13. New assets**

379 Answer: [Yes]

380 Justification: The LoRA adapter and training data will be released with appropriate docu-
381 mentation and licensing information.

382 **14. Crowdsourcing and research with human subjects**

383 Answer: [NA]

384 Justification: This work does not involve crowdsourcing or human subjects research.

385 **15. Institutional review board (IRB) approvals**

386 Answer: [NA]

387 Justification: No human subjects research was conducted.

388 **16. Declaration of LLM usage**

389 Answer: [Yes]

390 Justification: GPT-4o-mini is used as a core component of the method for generating reflection
391 training data, as described in Section 3.2.

A Training details

Hardware. All experiments were conducted on a Yale HPC (McCleary) cluster node equipped with a single NVIDIA A5000 GPU (24GB VRAM), 4 CPU cores, and 32GB system RAM.

Quantization configuration. We use BitsAndBytes 4-bit quantization with the following settings: NF4 quantization type, double quantization enabled, BF16 compute dtype. The quantized base model occupies approximately 5GB of GPU memory.

Training dynamics. Training loss decreases from 2.29 (step 5) to 0.25 (step 185), with training accuracy increasing from 0.60 to 0.94 over 189 training steps.

Safety taxonomy. The safety instruction prompt includes 13 categories: S1 (Violent Crimes), S2 (Non-Violent Crimes), S3 (Sex-Related Crimes), S4 (Child Sexual Exploitation), S5 (Defamation), S6 (Specialized Advice), S7 (Privacy), S8 (Intellectual Property), S9 (Indiscriminate Weapons), S10 (Hate), S11 (Suicide & Self-Harm), S12 (Sexual Content), S13 (Elections).

Reflection generation prompt (teacher). The GPT-4o-mini teacher model receives the following system prompt for generating reflection training data:

You are a safety analysis expert building training data for an AI safety classifier. For a given user message and its ground-truth safety label, write a concise logical self-reflection (2–4 sentences) that: (1) Identifies whether the message uses adversarial techniques to bypass safety filters (e.g., roleplay framing, fictional scenarios, indirect harmful requests, technical obfuscation, authority appeals, social engineering). (2) States whether the underlying intent is harmful or benign and why. (3) Lists the key indicators that support the classification. Rules: Be precise and analytical. Do NOT moralize or preach. For harmful prompts: explain the harm category and any adversarial technique used. For benign prompts: confirm safety and explain why. Output ONLY the reflection text (2–4 sentences). No preamble, no verdict.

B Additional error analysis

Table 5 shows the distribution of Reflect-Guard’s false negatives by harm subcategory.

Table 5: Reflect-Guard false negative distribution by harm subcategory on WildGuardTest.

Subcategory	Count
Private information (individual)	32
Social stereotypes & discrimination	26
Others	16
Fraud / assisting illegal activities	7
Mental health / over-reliance / crisis	6
Sensitive information (org/gov)	6
Copyright violations	5
Sexual content	4
Defamation / unethical actions	1
Total	103